

# A review on MPPT techniques of PV system under partial shading condition

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**Abstract**—Maximizing power output from solar photovoltaic (PV) systems is imperative for efficient utilization of renewable energy resources. However, partial shading, caused by factors such as cloud cover, foliage, or nearby structures, significantly impacts the performance of PV arrays, leading to suboptimal energy harvest. In this paper, we present an excellent review of Proportional Ratio Test (MPPT) technology, specifically designed to solve the problems caused by partial shading on the sun. We analyze and compare various MPPT algorithms, including traditional interference and observation (PO), incremental conductance (IncCond), and landscape (HC), as well as technology technologies such as fuzzy logic control, artificial neural network (ANN), and evolution. Additionally, we study a hybrid MPPT technique that combines various techniques to improve performance and robustness under different shading conditions. We also discuss practical issues and technological advances in MPPT hardware and software solutions. The purpose of our review is to provide researchers, engineers, and solar professionals with recommendations for the latest MPPT technologies to reduce the disadvantages of partial shading and increase the energy production of photovoltaic systems.

**Index Terms**—PV cell, Partial shading, MPPT

## I. INTRODUCTION

Due to the limited availability and increasing costs of traditional energy sources like coal and petroleum, along with their detrimental environmental effects, there is a growing push to meet energy needs through renewable sources such as solar, wind, hydro, and geothermal power for electricity generation [1]. Among these alternatives, solar or photovoltaic (PV) systems have gained significant popularity [2][3]. This is attributed to several advantages, including:

- Solar energy is abundant and free of fuel costs.
- PV systems generate no pollution or waste.
- They require less maintenance compared to other energy generation methods.
- These systems can operate unattended with minimal periodic maintenance, reducing labor costs.
- While initial installation costs may be Rapid, PV systems are cost-effective over the long term.
- Solar power can be generated locally, eliminating the need for long-distance transmission lines.

Solar energy systems can function as standalone units or be integrated with the grid, depending on grid availability [4][5][6]. This makes them particularly valuable for rural areas where grid infrastructure is sparse. Another advantage of using solar energy is the portable operation whenever wherever necessary. The constant increase in the development of the solar cells manufacturing technology would definitely

make the use of these technologies possible on a wider basis than what the scenario is presently. Photovoltaic (PV) cells are typically classified into three types: monocrystalline, polycrystalline, and thin-film. While PV cells generate current when exposed to sunlight, the amount of current produced is usually too small for practical use. PV cells are connected in series and parallel to produce the required voltage and current for operating a load. Thus, a cell is connected in series and parallel to form a PV module having a power rating in watts. The generated PV high voltages can be fed into power inverters for instance for use in the kilowatt (kW) or megawatt range (MW) [7]. These strings may be connected in series and parallel to produce strings, which is depicted in Figure 1, forming these arrays. It is the combined output of the individual PV modules that determines the power output of the PV array. PV arrays of a desired power rating can be formed by selecting the right number and arrangement of modules. Since the PV cell is the main part of the system, accurate modeling of a PV cell is essential for an optimal design of a PV system. Hence it is difficult to predict cell performance and optimize system design without proper modeling. It aims at better understanding of similar electrical behaviour of the cell in different conditions through high level model design. It also helps maximize energy output, minimize losses, and increase efficiency across the board. Hence, dependable models are crucial for stand-alone and grid-connected PV systems. Different modeling techniques have been proposed to describe how PV cells behave when subjected to different environmental factors. These models assist engineers and researchers in developing PV systems that are more effective and cost-efficient in harnessing solar energy [8]. Since a cell is the primary component of a PV system, it is necessary to model a PV cell accurately so that the PV system can be designed efficiently. To date, extensive articles are concerned about modeling and analysis of PV modules, with the specific topology of distinct PV system connecting in isolated or grid connected systems.

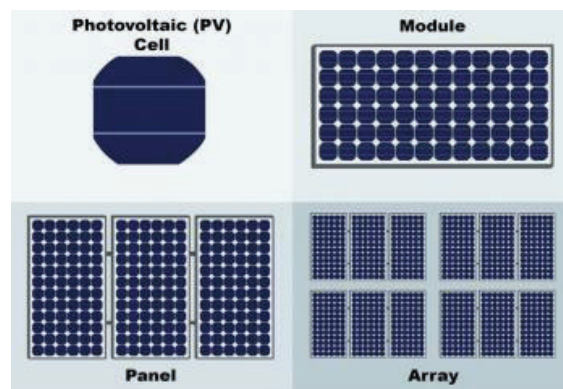


Fig. 1. Different configurations of PV cell

The topology selection is important for the success of the photovoltaic array structure. Various mathematical models of photovoltaic cells have been proposed in the literature [10][11][12][13][14][15], such as the ideal model, the double diode model, and the single diode model [9]. According to the laws of physics, an ideal photovoltaic (PV) module can be modeled as the current  $I_{ph}$  generated by sunlight in parallel with a diode. The diode represents the p-n junction of the photovoltaic module, and the current  $I_d$  flowing through the diode represents the current leaking from the p-n junction due to diffusion. This model is simple and non-destructive, but it does not reflect the real behavior of PV modules. To increase the accuracy, a resistance factor (RS) was added to account for the drop as shown in Figure 2. As shown in Figure 2[16][17]. To increase the accuracy of the model, a second diode was added to create a two-diode model. In this model,  $I_{d1}$  represents the diffusion current of the first carrier from diode D1, while  $I_{d2}$  represents the recombination current of the other carriers from diode D2. Although the two-diode model better describes the physical behavior of the photovoltaic module, it is nonlinear and more complex, which makes its mathematical investigation difficult [18] [19]. Usually, manufacturers provide parameters such as short-circuit current ( $I_{sc}$ ), open-circuit voltage ( $V_{oc}$ ), maximum power point voltage (VMPP) and maximum power point current (IMPP) in the data sheet. However, parameters such as photocurrent ( $I_{ph}$ ), saturation current ( $I_0$ ), series resistance ( $R_s$ ), shunt resistance ( $R_{sh}$ ) and diode ideality factor ( $a$ ) are not specified, which will cause users to be unknown. Therefore, the first step in the design of photovoltaic (PV) panels is to determine these parameters. In order to achieve a good design, it is important to provide these values correctly before the panel design. Although photovoltaic energy conversion is a rapidly developing technology in many countries, there are still challenges such as high installation costs, low power conversion and changes in energy production due to the environment. The output of photovoltaic (PV) systems is non-linear and their output is greatly affected by changes in solar radiation and temperature. Therefore, photovoltaic power changes greatly depending on the changing environment. In particular, the maximum power point (MPP) of photovoltaic cells changes significantly with changes in radiation and temperature [21]. To solve this problem, Maximum Power Point Tracking (MPPT) algorithm was developed [2] [22], which helps to increase the performance of solar modules and improve the use of renewable energy. Extensive research focuses on improving the efficiency and energy efficiency of photovoltaic systems [4][23][24]. One of the main problems of photovoltaic systems is the low voltage conversion caused by the inconsistency and time difference of current-voltage (I-V) and power-voltage (P-V) characteristics in response to irradiance and temperature changes. To maximize efficiency, PV systems should operate at their MPP to deliver maximum power [25] [26]. This is achieved by using an MPPT controller, which continuously monitors the MPP and adjusts the system to improve power production. Therefore, MPPT is considered as an important part of any photovoltaic system as shown in the MPPT block diagram (Figure 3).

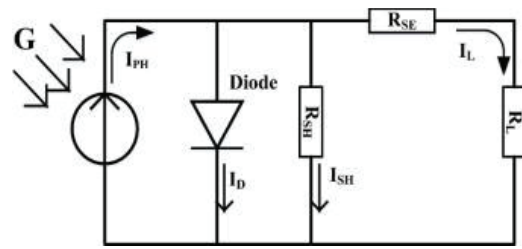


Fig. 2. Equivalent circuit of PV cell

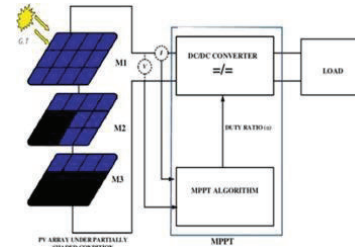


Fig. 3. MPPT block Diagram under PSCs

## II. LITERATURE REVIEW

Maximizing the energy output of solar photovoltaic (PV) systems is important to increase their efficiency and economic productivity. However, partial shading is a phenomenon that can affect the performance of optoelectronic systems, causing power loss and reducing the overall performance. In order to reduce the effects of partial shading, researchers have developed multiple maximum power output (MPPT) technology to improve the energy harvested by photovoltaic modules. This literature review investigates different MPPT techniques such as Probe Analysis (PO), Optimization (IncCond), Incremental Boost (INR), Two-Part Analysis Search Theorem (BST), Temperature Based MPPT, Variable Inductor MPPT, Fuzzy Logic, Artificial Neural Network (ANN) and Hybrid MPPT are designed to solve the partial competition problem of solar photovoltaics. Perturbation Observation (PO): Research results show that PO is a widely used MPPT technique due to its simplicity and efficiency. However, in some shades, PO may oscillate close to and deep into the maximum power point (MPP). Al-Nabulsi et al. (2018)

[34] proposed an improved PO algorithm (MPO) that performs better compared to the PO model to improve tracking accuracy based on partial shadows. Incremental Conductance (IncCond): Studies comparing IncCond with PO have shown that IncCond provides faster and more accurate MPP tracking, especially in shaded areas. Al-Hosani et al. (2018)[36] made a comparison between IncCond and PO in partial shade, demonstrating the superiority of IncCond in terms of tracking speed and robustness. Differences in step incremental resistance (INR): Ahmed et al. (2016) [37] proposed a novel adaptive MPPT algorithm based on INR to improve tracking performance in partial shade. This algorithm dynamically adjusts the step size of the incremental resistance to optimize MPP search, thereby reducing power loss and improving system performance. Two-part search theorem (BST): Marciantie et al. (2009)[38] introduced a BST method for MPPT control of photovoltaic modules under non-uniform irradiation conditions. The BST algorithm effectively identifies MPPs by minimizing the search time and performs well in the partial shade condition. Temperature measurement based on MPPT: Research results show that temperature measurement based on the MPPT algorithm improves the

tracking accuracy by integrating temperature changes into the control process. Lacey et al. (2014) [39] proposed a new method based on Simultaneous Perturbation Stochastic Approximation (SPSA) to improve MPPT under partial shading conditions by taking into account the irradiance and temperature variation. Variable inductor MPPT: Ali et al. (2017) [40] proposed a new variable inductor based MPPT algorithm suitable for photovoltaic systems in partial shade. By dynamically adjusting the inductance of the MPPT circuit, this algorithm effectively reduces the parasitic effect caused by shading and improves the overall performance of the system. Fuzzy logic: Abaza et al. (2019) [41] developed an improved fuzzy logic control MPPT method for optoelectronic systems under partial shading conditions. The fuzzy logic controller adaptively adjusts the operating parameters according to the shadow intensity and pattern, which shows increased search efficiency compared to traditional methods. Artificial Neural Networks (ANN): Mohamed et al. (2016) [42] proposed an MPPT algorithm for optoelectronic systems in partial shading based on artificial neural networks. The ANN model learns from historical data to accurately predict the MPP based on shadow change, providing better performance and flexibility compared to traditional methods. Hybrid MPPT: Raghavendra et al. (2019)[43] presented an optimized hybrid algorithm combining genetic methods for POPT in some shaded PV systems. This hybrid approach effectively combines the advantages of both algorithms to achieve improved tracking performance and robustness under dynamic shading conditions. Genetic Algorithm (GA): GA optimizes MPPT by adopting the natural selection process. Liu et al. (2018)[30] proposed a multi-objective MPPT algorithm based on genetic algorithm to improve power harvesting and system reliability. Find a table: A table finding method uses pre-calculated MPPs in a table to quickly determine the best workplace. Zhang et al. (2020)[63] studied the scalability of MPPT lookup table for large photovoltaic systems and proposed its improvement Differential evolution (DE): Zhao et al. (2020)[64] proposed a hybrid DE algorithm incorporating local search techniques to improve the performance for MPPT. DE is a population-based optimization algorithm for MPPT in photovoltaic systems. Particle Swarm Optimization (PSO): S. Mirjalili et

al. PSO optimizes MPPT by simulating the social behavior of the population. Voltage Window Search: The search window voltage adjusts the operating voltage in the window before finding the MPP. S. Tiwari et al. (2019) [53] performed simulations to evaluate the efficiency of electric window detection under dynamic shading conditions. State Space Based MPPT: State space based MPPT models PV system dynamics in state space framework for optimum MPP determination. K. Prabha et al. Split Rectangle: Split Rectangle method splits the I-V curve and searches for MPPs in each region. K.

K. Rajasekaran et al (2016)[49] investigated the application of Dividing Rectangles in large-scale PV systems, highlighting its scalability and robustness. Load Line MPPT: M. Belhocine et al (2018)[47] conducted simulations to evaluate the effectiveness of Load Line MPPT under various shading conditions. The Load Line MPPT method adjusts the load resistance to keep the operating point on the load line that corresponds to the maximum power point (MPP). The Fractional Voc or Isc method estimates the MPP by observing the fractional variation in the open-circuit voltage (Voc) or short-circuit current (Isc) caused by changes in shading. Y. C. Han et al (2015)[45] conducted experiments to validate the effectiveness of Fractional Isc- based MPPT algorithms in real-world scenarios. Maintaining the Integrity of the specification. M. S. Kumari et al (2020) proposed a hybrid DE algorithm for MPPT, integrating local search strategies for improved performance Swarm Optimization (PSO).

### III. COMPARISON OF DIFFERENT MPPT TECHNIQUES

To optimize MPPT methods, they must efficiently track the global MPP with high speed, low implementation complexity, independence from specific systems, and minimal cost. Table

1 outlines the advantages and drawbacks of each method. While some methods excel in tracking the global MPP under partial shading conditions, they often suffer from increased complexity or slower tracking speeds. Conversely, other MPPT techniques are effective only under uniform irradiance conditions.

TABLE I. COMPARISON OF DIFFERENT MPPT TECHNIQUES

| S.No | Max. Power Point Technique                      | Global Max. Power | Dependency of PV system | Accuracy of Tracking | Speed of Transient tracking | Level of complexity | Sensors required | Factors influencing performance                                                  |
|------|-------------------------------------------------|-------------------|-------------------------|----------------------|-----------------------------|---------------------|------------------|----------------------------------------------------------------------------------|
| 1    | Perturb and observe                             | Negative          | Negative                | Moderate             | Moderate                    | Least intricate     | V, I             | Rapid changing environmental conditions, step size, Partial shading              |
| 2    | Incremental conductance                         | Negative          | Negative                | Moderate             | Moderate                    | Moderate            | V, I             | Rapid changing environmental conditions, step size, Partial shading              |
| 3    | Fractional Voc or Isc                           | Negative          | Affirmative             | Reduced              | Rapid                       | Least intricate     | V or I           | PV pannel aging, irradiance variation, partial shading                           |
| 4    | Bisection search theorem (BST)                  | Negative          | Negative                | Elevated             | Moderate                    | Moderate            | V,I              | Partial shading                                                                  |
| 5    | Variable step size incremental resistance (INR) | Negative          | Negative                | Extremely elevated   | Instantaneous               | Intricate           | V, I             | Partial shading                                                                  |
| 6    | Temperature based MPPT                          | Negative          | Affirmative             | Reduced              | Moderate                    | Least intricate     | T, V             | Rapid changing irradiance, Partial Shading, Non uniform temperature distribution |
| 7    | Variable inductor MPPT                          | Negative          | Affirmative             | Moderate             | Moderate                    | Moderate            | V, I             | Reduced inductor size, Partial shading                                           |
| 8    | Load Line MPPT                                  | Partial y         | Negative                | Moderate             | Moderate                    | Moderate            | V, I             | PV pannel aging, partial shading                                                 |
| 9    | Dividing Rectangles                             | Partial y         | Negative                | Moderate             | Moderate                    | Intricate           | V,I              | Choice of control parameters                                                     |



|    |                                   |                                        |             |             |                    |               |                  |        |                                                                      |
|----|-----------------------------------|----------------------------------------|-------------|-------------|--------------------|---------------|------------------|--------|----------------------------------------------------------------------|
| 10 | State space based MPPT            |                                        | Affirmative | Affirmative | Elevated           | Rapid         | Highly Intricate | V, I   | Rapid changing irradiance, partial shading, complex design procedure |
| 11 | Voltage Window search             |                                        | Affirmative | Negative    | Moderate           | Moderate      | Intricate        | V, I   | Rapid changing environment, Initial working point                    |
| 12 | Evolutionary Algorithm based MPPT | PSO(Particle Swarm Optimization)       | Affirmative | Negative    | Extremely elevated | Rapid         | Moderate         | V,I    | Limited speed                                                        |
|    |                                   | DE(Differential Evolution)             | Affirmative | Negative    | Extremely elevated | Rapid         | Moderate         | V,I    | Limited speed                                                        |
|    |                                   | GA(Genetic Algorithm)                  | Affirmative | Negative    | Extremely elevated | Delayed       | Moderate         | V,I    | Slow Tracking speed                                                  |
| 13 | Fuzzy Logic                       |                                        | Affirmative | Affirmative | Extremely elevated | Rapid         | Intricate        | V, I   | Requirement of experience based fuzzy rule                           |
| 14 | Look up table                     |                                        | Affirmative | Affirmative | Reduced            | Rapid         | Moderate         | G, V,I | PV Pannel aging, system dependance                                   |
| 15 | ANN                               |                                        | Affirmative | Affirmative | Extremely elevated | Instantaneous | Moderate         | G, T   | Requirement of training data base                                    |
| 16 | Hybrid MPPT                       | Model Heuristic Technique              | Negative    | Affirmative | Extremely elevated | Moderate      | Intricate        | V,I    | Requirement of System Model, PV array aging                          |
|    |                                   | RCC + Model Reference Adaptive Control | Negative    | Affirmative | Extremely elevated | Rapid         | Intricate        | V,I    | Requirement of System Model, PV array aging                          |
|    |                                   | ANN + Fuzzy                            | Negative    | Affirmative | Extremely elevated | Instantaneous | Intricate        | G,T,V  | Requirement of Training Data and Fuzzy Rules                         |

#### IV. CONCLUSION

This paper explores various methods for modeling photovoltaic (PV) arrays and maximizing their power output through maximum power point tracking (MPPT). It discusses the limitations of current models which often assume static parameters like insolation and temperature, leading to inaccuracies in real-world conditions where these variables fluctuate significantly. This poses challenges for accurately predicting PV array performance.

Intelligent modeling techniques, such as those employing artificial intelligence (AI), are capable of more precise predictions but often at the cost of increased computational complexity. Thus, there is a demand for modeling approaches that strike a balance between accuracy and computational efficiency.

MPPT methods are crucial for optimizing PV array performance by swiftly identifying and tracking the global maximum power point under varying environmental conditions. However, existing methods exhibit varying degrees of effectiveness in handling scenarios like partial shading and rapid changes in irradiance, which are common in tropical regions.

In summary, this paper highlights the urgent need for enhanced modeling techniques that better reflect real-world conditions and MPPT strategies that are adept at adapting to dynamic environmental factors, particularly in challenging geographic locations.

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